From Point Forecasts to Distributions: Adapting TimesFM to Synth

SYNTH

www.synthdata.co

July 2026

Abstract

Synth (Bittensor Subnet 50) is a forecasting competition. Each miner submits not a single prediction but a *spread* of possible future price paths for a set of assets, and is scored on how well that spread matched what actually happened. Being right on average is not enough: you are graded on your uncertainty.

This note asks a practical question. Take a strong off-the-shelf forecasting model—TimesFM 2.5 [1, 2]—and make it competitive on Synth: how hard is it, and how far can you get? We answer as a ladder of seven steps and measure each one against the live field.

Using only the raw point forecast—replicated across every path—lands near the bottom of the ∼256-miner field. **One cheap step does most of the work**—sampling the model's own built-in ranges instead of its single prediction (mean rank ∼230 → ∼188, error down ∼19%, no training, no GPU). Tail completion, drift re-centering, and a 15-minute fine-tune each improve the multi-window average; per-asset tuning is effectively null. The gains after R2 are modest and their confidence intervals overlap. The full ladder reaches mean rank ∼167, with the fine-tune the best and most stable configuration. The whole thing cost about three days and under an hour of GPU on a gaming graphics card. We do *not* beat our own deployed Chronos-2 miner [3]—one of the field's strongest—at the 24-hour horizon (∼167 vs. ∼91), and were not trying to; at 1 hour TimesFM is slightly ahead on the point estimates, but with overlapping confidence intervals the comparison is inconclusive. Everything is measured across nine (24-hour) and eight (1-hour) regime windows with bootstrap confidence intervals.

1 Introduction

Synth [7, 6] is a benchmark and live competition for probabilistic price forecasting, run as Bittensor Subnet 50. It asks each miner a specific question: not "where will the price be?" but "what is the range of paths the price might take?" For each asset the miner submits an ensemble of possible future trajectories. The validator scores that ensemble against what actually happened using CRPS [5], a proper scoring rule that rewards forecasts that are both accurate *and* honestly spread. A confident-but-wrong forecast loses to a slightly-off-but- correctly-spread one.

That is a problem for today's time-series foundation models. Most are built to output a *point* forecast—a single best-guess path—which is the wrong shape for Synth. TimesFM 2.5 [1, 2] is one of the strongest. Its native outputs are a point forecast plus a set of per-step ranges (marginal quantiles). Submit the point forecast directly—every path identical, zero spread—and it ranks about 224th of ∼256 miners on our test window. The model is good; the *object* it emits is the wrong one for the game.

So how good can an adapted TimesFM get, and how much work is it? For a yardstick we use our own live Chronos-2 miner [3], one of the strongest on the subnet. Under the same strict backtest protocol we apply to every TimesFM arm—per-prompt error rank against the live field, averaged over the 12 assets and across windows—Chronos-2 averages rank about 91 at 24 hours, versus roughly 230 for the point-only TimesFM baseline. (That rank of 91 is worse than its smoothed leaderboard standing, because the strict protocol includes every asset and every day; it is simply the honest, like-for-like bar.) This note documents, step by step, how far a few days of adaptation closes that gap, measured across many market regimes rather than one lucky (or unlucky) window. Each step—we call them *rungs*—isolates one design decision.

2 The model: TimesFM 2.5

TimesFM 2.5 [1, 2] is a ∼200M-parameter foundation model for time series. It is pretrained on a huge mix of series and runs on a new one with no extra training. We feed it the recent history of returns,

$$r_t = \ln(P_t/P_{t-1}), \tag{1}$$

and it forecasts the next stretch. Out of the box it gives two things per future step: a single best-guess value (the point forecast) and nine ranges around it—the deciles $q_{10}, \ldots, q_{90}$. Two facts about this output shape everything below.

- **It describes each step on its own, not whole paths.** The deciles tell you the likely range at step 5, at step 10, and so on—but not how those steps join into a single trajectory. Synth scores whole paths, so we have to build the path-sampling ourselves.
- **The deciles are as wide as this setup sees.** We used the default nine deciles and asked for nothing past q_{10}/q_{90}, so the model says nothing about the outer 20% of its own distribution—exactly the big-move tails CRPS cares most about. (TimesFM 2.5 can expose finer quantiles in principle [2]; this is a limit of our setup, not the model.) An earlier quick test of ours failed on precisely these clipped tails, so we add a rung to fix it.

3 The adaptation ladder

Each rung adds one idea on top of the last. They all use the same inputs (the trailing week of minute prices—no external data, no peeking ahead) and the same scoring.

R0 — the point forecast. Every path in the ensemble is the same single best-guess line. This is a deliberately dumb baseline: zero spread, which CRPS punishes hard. It ranks near the bottom (∼215–238 across windows). The model has quantiles too—R0 just throws them away, to show what emitting no uncertainty costs.

R1 — point + Gaussian noise. A newcomer's first instinct: keep the best-guess line and scatter noise around it, sized to how volatile the asset has been lately. Better than R0 in most windows. But the noise is one fixed number, blind to context: it over-spreads a stock overnight and on weekends, and in calm markets can score *worse* than the plain line.

R2 — the native decile head. Drop the hand-made noise. Instead, draw each step's move from the model's *own* nine ranges (the deciles). Now the spread comes from the forecaster: it tightens and widens with the time of day, the market regime, the asset. **This is the biggest and most reliable jump on the whole ladder** (Section 3.1)—and it needs no training and no GPU beyond a normal forecast.

R3 — tail completion beyond the deciles. The deciles stop at q_{10}/q_{90}, so R2 says nothing about the extremes. We graft a tail past them: a bounded, Student-t-shaped extension joined smoothly to the decile band. This fills in the big-move tails CRPS weighs heavily, and directly targets the clipped-tails failure from Section 2.

R4 — directional-drift re-centering. Fed a recent trend, the model tends to keep extrapolating it—a built-in directional bet. We shift the ensemble so its center carries no such bet, keeping the spread unchanged. Our Chronos-2 miner [3] learned the same lesson the hard way. It helps most on trending crypto.

R5 — per-asset tuning (≈ null, and that is a finding). Try tuning the re-centering strength and tail thickness *per asset*, chosen across three different market regimes. Result: nothing survives for 11 of 12 assets; the one accepted change is worth ∼0.3 ranks. Per-asset "signals" from any single window turned out to be luck. We keep the null because it is the finding: the ladder's gains are structural, not fine tuning. The error (CRPS) agrees—R5 and R4 are within noise (Table 1).

R6 — distributional LoRA fine-tune. Finally we train. Small LoRA adapters [4] nudge the weights, using twelve months of past returns that end before every evaluation window (cutoff 2026-05-24). A checkpoint takes about fifteen minutes on a gaming GPU (RTX 3070). Across the nine regime windows it is the best and most stable adapted configuration (Section 3.1)—the clearest single sign that, once the distribution is built and refined (R2–R4), a lightweight fine-tune still adds value.

3.1 Measured ladder: nine regime windows

We measure two things per rung: its estimated rank against the live field (out of ∼256, lower better) and its weighted CRPS (the raw error, lower better; "coef-weighted" = assets combined with the subnet's own per-asset weights). We evaluate on **nine two-day windows spanning June to mid-July 2026**, deliberately spread across regimes—uptrends, downtrends, calm stretches, a high-volatility spell, and chop—all after the fine-tune's training cutoff and none used to tune anything. Reporting many windows rather than one is the point: it gives a mean *and* a 95% bootstrap confidence interval across regimes. It also matters here specifically, because an earlier read on a single short window had ranked the fine-tune dead last; the multi-regime picture below reverses that. Our live Chronos-2 miner [3] is the reference line and the anchor for the rank estimate (Section 5).

Table 1: 24-hour profile (LOW): mean estimated rank and mean weighted CRPS across the nine regime windows, with 95% bootstrap confidence intervals (resampling windows). Lower is better; field ∼256. Chronos-2 is the live reference miner.

Arm	mean rank [95% CI]	mean CRPS [95% CI]
Chronos-2 (reference, live)	91 [82, 105]	2264 [1752, 2790]
R0 point	230 [226, 235]	2965 [2283, 3673]
R1 + Gaussian noise	209 [191, 225]	2546 [2098, 3009]
R2 decile sampler	188 [167, 207]	2406 [1853, 2976]
R3 + tail completion	178 [161, 194]	2371 [1839, 2921]
R4 + drift re-center	174 [155, 192]	2356 [1828, 2899]
R5 + per-asset tuning	174 [155, 192]	2356 [1828, 2899]
R6 + LoRA fine-tune	**167 [158, 177]**	**2324 [1805, 2857]**

Reading the ladder (Table 1, Figure 1):

1. **The big win is nearly free: use the model's own ranges.** R0→R2 is by far the largest step—mean rank 230 → 188, error down ∼19%—with no training and no GPU beyond a normal forecast. Just treat the built-in quantiles as a real distribution.

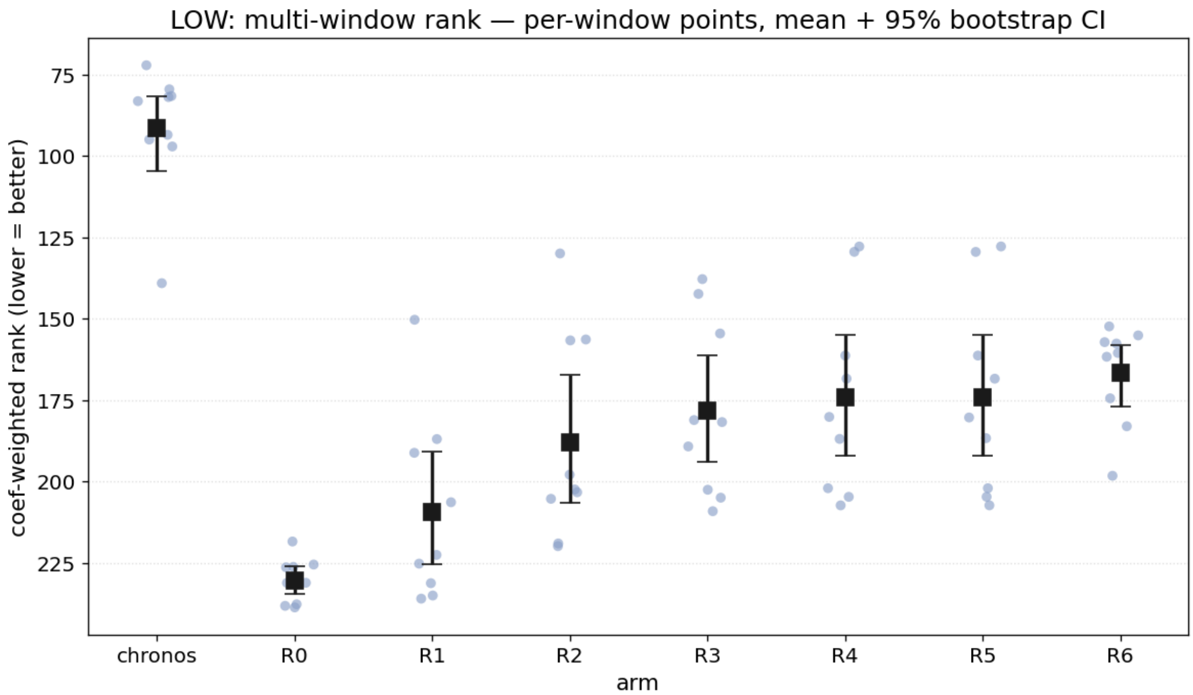

Figure 1: The 24-hour ladder across nine regime windows. Per rung: the per-window estimated rank (points), the mean (marker) and its 95% bootstrap CI (bar); the Chronos-2 reference band spans its per-window range. Lower is better. Performance improves from R0 to R4, is unchanged at R5, and reaches its best mean and tightest spread at R6 (the fine-tune).

2. **Most later steps help too—except per-asset tuning.** Tail completion, drift re-centering, and fine-tuning each improve the multi-window average (188 → 178 → 174, then 167 at R6 in rank; 2406 → 2324 in CRPS); per-asset tuning (R5) is effectively null (see below). The fine-tune (R6) is the **best adapted configuration** (167, CI 158–177) and, tellingly, the **most stable across regimes**—its window-to-window rank stays in a tight 152–183 band where the plain R2 sampler swings from 130 to 220. The gains past R2 are each smaller than the R0→R2 jump and their CIs overlap, so we read them as real but modest. (This overturns an earlier single-window read that had R6 worst and R2 best—a good reminder of why multiple regime windows matter.)

3. **Per-asset tuning (R5) is null.** R5 equals R4 to within noise on every window; static per-asset tuning bought nothing.

Bottom line for 24 hours: adapted TimesFM improves steadily but stays clearly behind our Chronos-2 miner (mean rank 167 vs. 91; error 2324 vs. 2264). The ladder was the deliverable, not a dethroning.

3.2 The 1-hour horizon: TimesFM is competitive with Chronos-2

We ran the same ladder at Synth's 1-hour crypto horizon (1-minute bars, five assets) over eight one-day regime windows (Table 2, Figure 2). Two results:

- **Here TimesFM edges ahead of our Chronos-2 miner.** The best adapted arm—a fine-tuned decile sampler (ft-R2)—averages rank 116 (CI 94–132) vs. Chronos-2's 130, and leads on error too (848 vs. 872). Several TimesFM arms beat the reference on the point estimate. But the confidence intervals are wide and overlapping, so the advantage remains *uncertain*—TimesFM is slightly ahead on average, not a clear win—and it is helped by the 1-hour profile being Chronos-2's weaker leg.

- **The 24-hour levers partly invert.** Tail completion (R3) *hurts* at 1 hour and the drift re-center (R4) is roughly neutral: at this short horizon the native band is already adequate, echoing what we found on Chronos-2, where band-widening tricks also died short. But the fine-tune—which a single earlier window had made look useless at 1 hour—in fact *transfers positively*: the two best arms are both fine-tuned.

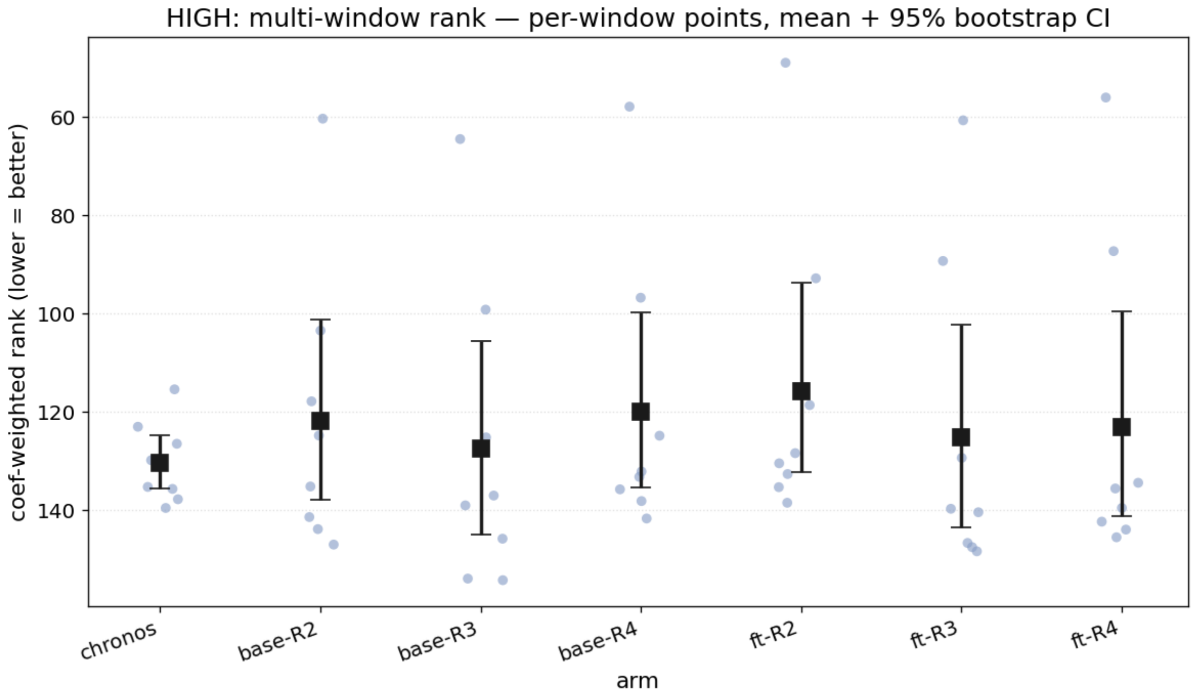

Figure 2: The 1-hour study across eight regime windows: six arms (pretrained "base" and LoRA-"ft" weights × rungs R2/R3/R4) plus the Chronos-2 reference. Per-window points, mean and 95% bootstrap CI. Lower is better; the fine-tuned decile sampler (ft-R2) is best on average.

Table 2: 1-hour profile (HIGH): mean estimated rank and mean weighted CRPS across eight regime windows, 95% bootstrap CIs. "base" = pretrained weights, "ft" = LoRA fine-tuned on 5-minute returns. Lower is better.

Arm	mean rank [95% CI]	mean CRPS [95% CI]
Chronos-2 (reference, live)	130 [125, 135]	872 [688, 1062]
base-R2	122 [101, 138]	882 [655, 1108]
base-R3	127 [106, 145]	867 [644, 1089]
base-R4	120 [100, 135]	838 [621, 1053]
ft-R2	**116 [94, 132]**	848 [627, 1066]
ft-R3	125 [102, 143]	843 [623, 1059]
ft-R4	123 [100, 141]	837 [618, 1053]

4 Output-distribution characterization

What do the distributions we emit actually look like? The figures below profile the shipped 24-hour setup over ∼200 recent prompts per asset (100 paths each). These are properties of what the model puts out—nothing here is scored against reality.

4.1 Volatility follows the trading day

Stocks and oil show a sharp jump in predicted spread at the NYSE open (∼14:30 UTC), staying wide through the US session and quieting overnight (Figure 3); gold is mild, and crypto shows a faint version of the same bump on an otherwise flat 24/7 profile. We hard-coded none of this: the model reads it off the past week of prices, and it flows straight into the sampled paths through the deciles. That is exactly the payoff of R2 over R1's blind, flat noise.

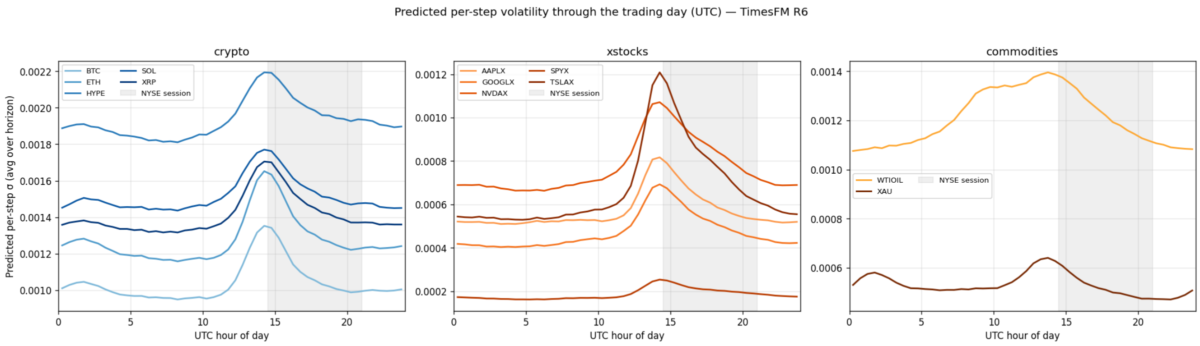

Figure 3: Predicted per-step σ vs. UTC hour, faceted by asset class; NYSE session shaded. Session-bound assets spike at the open and stay elevated through the US session; crypto shows a much milder bump on a flat profile.

4.2 Increment shape: a concentrated peak with truncated tails

Pool every sampled single-step move for an asset and standardize it: the result is mildly peaked (excess kurtosis ≈ 1.05 BTC, 1.65 SPYX, 0.70 WTIOIL; Figure 4). It matters *why* it is peaked. It is **not** fat tails: the moves stop dead beyond about ± 3.5–4.5σ, because the quantile grid (deciles plus the clamped tail) can only reach so far. The peakedness comes from a tall center and shoulders above the Gaussian, next to those cut-off extremes. In short the miner is "peaked and clipped," not "fat-tailed"—a direct fingerprint of the deciles-only setup from Section 2, and the clearest thing a finer quantile head would fix.

4.3 Cross-asset shape

Table 3 lists per-asset shape numbers. The spread ordering is sensible (HYPE widest, SPYX narrowest, a factor of ∼10). The tail ratio sits at 2.07–2.23 vs. 1.82 for a Gaussian—a bit heavier-shouldered—and the shapes are essentially symmetric. One honest note: the tail ratio is nearly *the same* for eleven of twelve assets (2.22–2.23). That sameness is our own tail shape showing through—past the deciles the geometry comes from our parametric fill, not from the model—whereas the widths, which the model does set per asset, vary strongly.

4.4 Horizon growth

The predicted spread grows like $\sqrt{t}$ (Figure 5)—the textbook random-walk rate, with every asset's curve running parallel to the reference. In plain terms, uncertainty widens with the horizon at the right pace: neither pinched in nor blowing up over 24 hours.

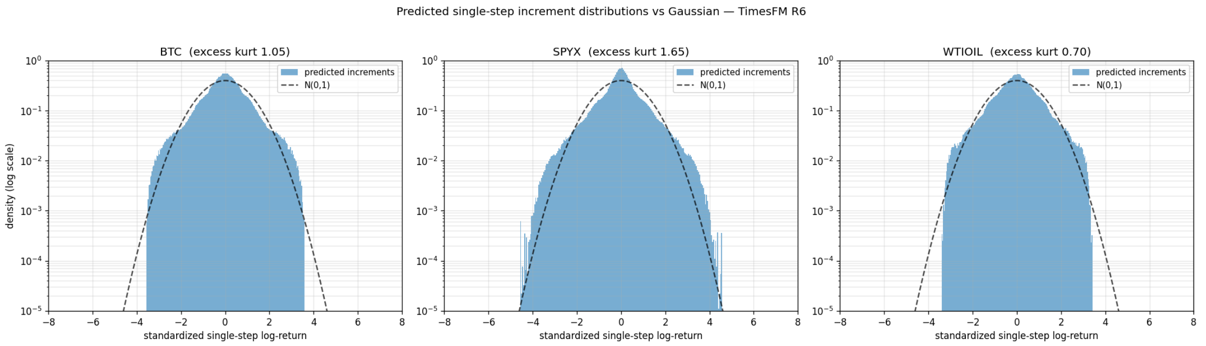

Figure 4: Pooled single-step log-returns (all forecast steps and paths), standardized to unit variance, log-density, with a matched $\mathcal{N}(0,1)$, for BTC, SPYX, WTIOIL. The excess kurtosis is produced by the concentrated peak and elevated shoulders; the tails are *truncated* at $\sim\pm 3.5$–4.5σ by the bounded quantile-grid sampler, not fat.

Table 3: Distributional shape statistics of the final 24-hour configuration. Per-step σ in basis points of log-return; tail and asymmetry ratios computed from the sampled ensemble, per step then averaged over steps and prompts. Gaussian reference: tail ratio = 1.82, symmetric asymmetry = 1.0.

Asset	σ_{step1} (bps)	σ_{final} (bps)	$(q_{99}-q_{01})/(q_{90}-q_{10})$	$(q_{90}-q_{50})/(q_{50}-q_{10})$
AAPLX	5.60	6.53	2.23	1.05
BTC	10.18	11.33	2.22	1.01
ETH	12.53	14.03	2.22	1.00
GOOGLX	4.78	5.41	2.23	1.00
HYPE	18.83	20.16	2.22	1.07
NVDAX	7.72	8.63	2.23	1.00
SOL	14.55	16.40	2.22	1.00
SPYX	1.87	2.09	2.23	0.98
TSLAX	6.47	7.50	2.23	1.00
WTIOIL	12.82	13.49	2.07	1.01
XAU	5.14	6.02	2.23	1.02
XRP	14.11	15.33	2.22	1.01

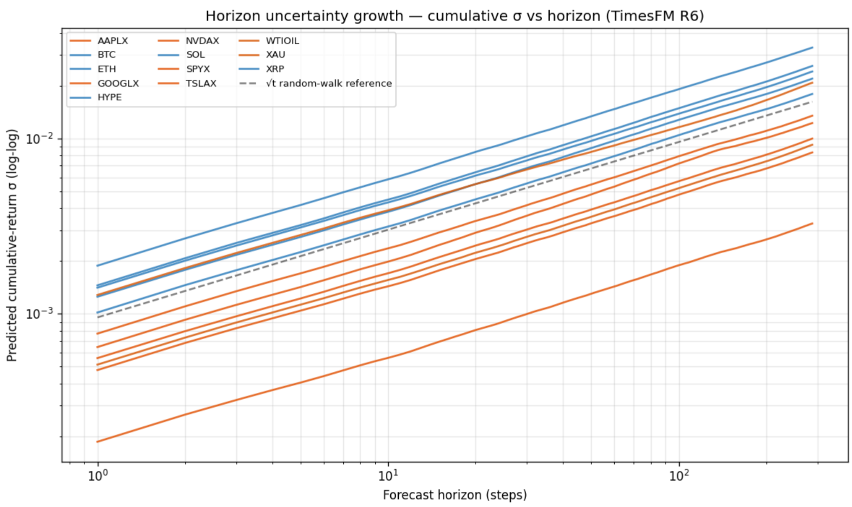

Figure 5: Predicted cumulative-return σ vs. horizon (log–log, 12 assets) with a $\sqrt{t}$ random-walk reference.

5 Evaluation protocol and estimated standing

One thing to be upfront about: every rank here is **estimated from backtests, not live**—no TimesFM setup has ever run on the subnet. To estimate a rank, we score each setup on past prompts and place its error inside the real validator scores of the live field for those same prompts, using our deployed Chronos-2 miner (which is in both datasets) to calibrate the mapping. The estimate is only as good as that calibration. We evaluate across nine 24-hour and eight 1-hour windows spanning June to mid-July 2026, chosen for regime variety, all after the fine-tune's 2026-05-24 training cutoff; the bootstrap CIs in every table resample those windows. Three caveats to carry into every table:

- **We start from the true price.** Our setups are anchored on the real starting price of each prompt; the live reference miner runs under real conditions (feed lag and all). So our rank-vs-Chronos gap flatters us a little. The *ordering within our ladder* is unaffected.
- **We use fewer paths.** 100 sample paths vs. the field's 1000—a handicap the other way, and one that can vary by setup and asset.
- **SPCX is left out on most windows.** It was listed too recently to have the seven days of history our method needs, so it is absent from every window before mid-July—a real data wall any newcomer serving a fresh asset will hit.

An earlier version of this note evaluated on a single short window; it happened to rank the fine-tune last and called the 1-hour result a fluke. Re-running across many regimes reversed both—the reason we now report windowed means with confidence intervals rather than a single number.

6 Effort: the accessibility argument

The whole thing—seven 24-hour rungs, the 1-hour study, the per-asset sweep, the fine-tune, and the seventeen regime windows behind every figure here—took **a few days** of work. Total GPU time was well under an hour, and everything (training and inference) ran on a single gaming graphics card (RTX 3070). No cluster, no paid data, no proprietary model. Going from "bottom of the field" to "inside the field's main body" is, in effort, small—and that is the whole point of publishing it: the barrier to entry on Synth is know-how, not a compute budget.

7 Limitations and directions

Four limits point to the work we think is worth doing next. *The winners depend on the regime*: tails, re-centering and fine-tuning each help or hurt with the market, at both horizons—so the next step is not one fixed setup but *picking the rung on the fly* from what the market is doing, and the ladder already gives us the pieces. *We are capped at the deciles*: our setup stops at q_{10}/q_{90} and fills the rest by hand (Section 4); asking TimesFM 2.5 for its finer quantiles—which it can give—would let the model shape its own tails instead. *Fresh assets like SPCX*: too new to backtest, they need a short-context mode or a cold-start rule. *We have not gone live*: everything here is a backtest, and turning a backtest into a real miner—feed lag, request bursts, deadlines—is its own project, as our Chronos-2 miner's history shows.

References

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